

04T: Prediction for SLR

Stat 230 – Spring 2026

Note

This is the tutorial version of the R activity. If you prefer to use .qmd and RStudio directly, you can use <https://aluby.github.io/stat230/activities/04R-prediction>

1 Loading data

The data set is loaded by running the following code chunk:

```
city <- readr::read_csv(url("https://raw.githubusercontent.com/zief0002/modeling/main/"))
```

The columns of interest are education and income.

2 Making plots

Q1. Make a plot of income against education and overlay the linear regression equation. Does it look like what you expected? (I've started the code for you; you will need to overlay the linear regression equation)

```
#l min-lines: 3
#l fig-alt: "Scatterplot with education on the x-axis and income on the y-axis."
gf_point(income ~ education, data = city)
```

Q2. Make a plot of income against `scale(education)` and overlay the linear regression equation. Does it look like what you expected?

```
#l min-lines: 3
# Write your graph code here
```

3 Fitting a simple linear regression model

Q3. Use the `lm()` command to fit the simple linear regression model where education is used to predict income. Make sure your output matches the output on the worksheet.

```
#l min-lines: 3
# Write your lm() command here
```

4 Making predictions

The first person in this dataset has `education = 8`.

Q4. Calculate the expected income of this person using the first regression equation. (You may use R or do this by hand. There's a code chunk below for your convenience)

```
#l min-lines: 3
# Write your code here
```

Q5. If we want to predict the income of **this** person, should we use a confidence interval or a prediction interval?

Q6. Use R to construct the appropriate 89% interval for this person's income.

```
#l min-lines: 3
# Write your code here
```

Q7. Interpret the interval in context.

5 Plotting our intervals

It can be useful to plot your confidence/prediction intervals to communicate your findings to a broader audience. The code below creates a scatterplot of the data, adds the fitted regression line, and then adds 89% prediction intervals.

```
#l fig-alt: "Scatterplot with education on the x-axis and income on the y-axis, with
gf_point(income ~ education, data = city) |>
  gf_lm(interval = "prediction", level = .89) |>
  gf_lm(interval = "confidence", alpha = 0.6, level = .89)
```

Q8. Run the code to produce a scatterplot, regression line, and both types of intervals. Which is the prediction interval and which is the confidence interval? How can you tell?

6 Function quick reference

The following table summarizes the functions we learned today:

Function	Purpose
<code>lm(formula, data)</code>	Fit a linear model
<code>predict(model, newdata, interval, level, se.fit)</code>	Make predictions and calculate intervals
<code>gf_point(formula, data) > gf_lm(interval = "confidence")</code>	Plot data and add confidence intervals for the mean response
<code>gf_point(formula, data) > gf_lm(interval = "prediction")</code>	Plot data and add prediction intervals for the mean response